Chrisnerg Mqobo

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PROFESSIONAL SUMMARY

Full-stack developer with production experience building real-time monitoring systems, WebSocket-driven state synchronization, and Windows OS-level tooling. Proficient in React, TypeScript, Node.js, Electron, and PostgreSQL. Completed BSc Computer Science and Informatics at the University of Johannesburg (ranked #1 in SA for Computer Science) in November 2025, awaiting graduation ceremony May 2026. Adept at writing clean, testable code and delivering reliable software in fast-paced team environments.

EXPERIENCE

Junior Software Engineer

Tontrac Systems Analytics **October 2025 - Present, Sandton, Gauteng**

- Architected a real-time camera monitoring dashboard using Electron and React, enabling site operators to track multiple checkpoint feeds simultaneously via a high-performance grid layout.
- Implemented a secure IPC communication layer (Main → Preload → Renderer bridge) with WebSocket (WS) contracts to synchronise device state updates from a Master Agent proxy in real time.
- Engineered low-level Windows OS integrations using the Windows Registry API (HKCU) and Win32 broadcasts (WM_SETTINGCHANGE) to automate system-wide date/time configuration, eliminating manual setup steps.
- Designed diagnostic modal workflows with sequential “Snapshot All” capture logic and real-time image analysis results, improving operator incident response efficiency.
- Built a Playwright end-to-end testing suite that reduced manual regression testing effort by 40% and identified critical edge cases prior to production deployment.
- Collaborated in an Agile team environment using Git and GitHub for version control and code reviews.

Informatics Tutor

University of Johannesburg **February 2025 - September 2025, Auckland Park, Johannesburg**

- Mentored 100+ undergraduate students in algorithmic thinking, debugging strategies, and clean coding practices in Visual Basic, contributing to measurable improvements in module pass rates.
- Co-facilitated technical workshops with academic staff to reinforce curriculum objectives and boost student engagement across core informatics modules.

PROJECT

House Marketplace (React Frontend)

- Built a dynamic property listing platform in React.js with advanced search filtering, interactive listing components, and responsive UI design.
- Applied modern React state management patterns (Zustand) to manage complex UI states and deliver a seamless user experience across listing and search flows.

URL Shortener (Full-Stack)

- Architected a full-stack URL shortening service with a Node.js/Express.js REST API backend and a React TypeScript frontend.
- Selected Drizzle ORM over Prisma to eliminate Rust-binary overhead and minimize bundle size, achieving better query performance through a closer-to-SQL abstraction layer.
- Integrated Zod schema validation and JWT-based authentication to prevent unauthorized URL manipulation and ensure type-safe API contracts.

EDUCATION

Bachelor of Science in Computer Science and Informatics

Minor in Mathematics, IT Management • University of Johannesburg | Studies completed: Nov 2025 | Ceremony: May 2026

- Johannesburg Auckland Park • 2026

SKILLS

Languages: JavaScript (ES6+), TypeScript, Java, Python, SQL | Familiar: C#, C++, Visual Basic.

Frameworks/Libraries: React, Next.js, Electron, Node.js, Express.js, Flask| Familiar: Spring Boot, ASP.NET

Tools & Databases: PostgreSQL, MongoDB, Git, GitHub, Docker, Playwright, Postman, CI/CD.

Specializations: WebSocket (WS), IPC Architecture, Windows Registry API, RESTful APIs, Zod, Agile, Scrum.

Soft Skills: Communication, Teamwork, Problem-Solving, Adaptability, Mentorship, Critical Thinking, Time Management.